MINOR PROJECT REPORT

Exploratory Data Analysis on Sales Dataset

Sales Data Analysis and Business Insights Using Python

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Project Type	Minor Project — Data Science
Domain	Data Science / Exploratory Data Analysis

Professional project documentation prepared according to the required submission structure.

DECLARATION

I, Singipurapu Dinesh, hereby declare that this Minor Project titled “Exploratory Data Analysis on Sales Dataset” has been completed individually as part of the Data Science internship requirements at Topgrade Innovation. The analysis, visualizations, observations, and project documentation presented in this report are based on the project dataset and the work performed in the accompanying Python/Jupyter Notebook. The screenshots included in this report represent the actual execution of the completed project workflow.

ACKNOWLEDGEMENT

I sincerely thank Topgrade Innovation for providing the internship opportunity, training, and practical guidance required to complete this Minor Project. I am thankful for the practical exposure that helped me apply Python and data science concepts to a sales analysis problem. I also acknowledge the Python ecosystem, particularly Pandas, NumPy, Matplotlib, and Seaborn, which supported the data preparation, analysis, and visualization stages.

ABSTRACT

This project presents an Exploratory Data Analysis (EDA) of a sales dataset containing Date, Product, Quantity, and Revenue fields. The main objective is to transform raw transaction records into useful business information through data loading, quality checking, cleaning, aggregation, visualization, and interpretation. The analysis examines missing values, product-wise performance, monthly revenue trends, revenue contribution, and sales volume. The final results show that Laptop is the leading revenue-generating product, Mouse has the highest quantity sold, May 2025 is the strongest revenue month, and November 2025 is the weakest revenue month.

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1. Introduction

Exploratory Data Analysis is an important step in the data science workflow. It helps identify patterns, trends, relationships, unusual values, and important business indicators before advanced modeling is attempted. In this project, sales transaction data is analyzed to understand product performance, sales volume, and monthly revenue trends.

2. Problem Statement

A raw sales dataset contains individual transactions, but the raw records do not immediately show which products perform best, which products sell the most units, or which months generate the highest revenue. The project addresses this by cleaning the data, summarizing it, visualizing the results, and producing meaningful business insights.

3. Objectives

- Load and inspect the sales dataset.
- Identify and handle missing values.
- Perform basic data cleaning.
- Calculate total and average revenue.
- Calculate total and average quantity sold.
- Analyze product-wise revenue and quantity.
- Analyze monthly revenue trends.
- Create bar, line, and pie charts.
- Write meaningful business observations.

4. Dataset Description

The dataset contains four required attributes: Date, Product, Quantity, and Revenue. Date represents the transaction date, Product identifies the item sold, Quantity represents the number of units sold, and Revenue represents the revenue generated.

Field	Description
Date	Date of the sales transaction
Product	Product sold
Quantity	Number of units sold
Revenue	Revenue generated

5. Tools and Technologies

Technology	Purpose
Python	Core programming language used for the analysis.
Jupyter Notebook	Interactive environment used to execute and document the analysis.
Pandas	Data loading, cleaning, grouping, aggregation, and tabular analysis.

NumPy	Numerical operations and statistical calculations.
Matplotlib	Chart creation and visualization.
Seaborn	Statistical visualization and enhanced plotting.

6. Project Methodology

1. Load the raw sales CSV using Pandas.
2. Inspect dataset structure, columns, data types, and initial records.
3. Check missing values and duplicates.
4. Clean required fields and prepare the data for analysis.
5. Calculate overall revenue and quantity metrics.
6. Generate product-wise and monthly summaries.
7. Create bar, line, and pie visualizations.
8. Interpret the results and generate business insights.
9. Present the final results and maintain screenshot evidence.

7. Data Loading and Initial Inspection

The dataset was successfully loaded using Pandas. The first five rows were displayed to verify the expected columns and values. The screenshot evidence in Section 13 shows the actual notebook execution and initial records.

8. Missing Value Analysis

The missing-value analysis identified no missing values in Date. Product, Quantity, and Revenue each contained one missing value, approximately 0.33% of the dataset. These missing values were handled during the cleaning process.

9. Data Cleaning

The cleaned dataset retained the required Date, Product, Quantity, and Revenue fields and was prepared for analysis after the data-quality preparation steps. The cleaned records were then used for product and monthly analysis.

10. Product-wise Analysis

Product-wise aggregation was performed to calculate total revenue and total quantity for each product. Laptop generated the highest revenue at ₹12,164,760.73, followed by Smartphone at ₹6,885,490.38 and Monitor at ₹2,805,942.35. Mouse recorded the highest quantity sold with 405 units.

11. Data Visualization

Three principal visualizations were used to communicate the findings clearly: a bar chart comparing revenue by product, a line chart showing monthly revenue trends, and a pie chart showing each product's share of total revenue.

11.1 Bar Chart — Revenue by Product

The bar chart compares total revenue by product. Laptop has a substantially taller bar than the other products, showing its dominant contribution to revenue.

11.2 Line Chart — Monthly Revenue Trend

The monthly line chart shows revenue changes across 2025. Revenue reaches its highest point in May and falls to its lowest point in November.

11.3 Pie Chart — Revenue Share by Product

The pie chart shows each product's contribution to total revenue. Laptop contributes approximately 50.0% and Smartphone approximately 28.3%; together they account for roughly 78.3% of total revenue.

12. Business Insights and Observations

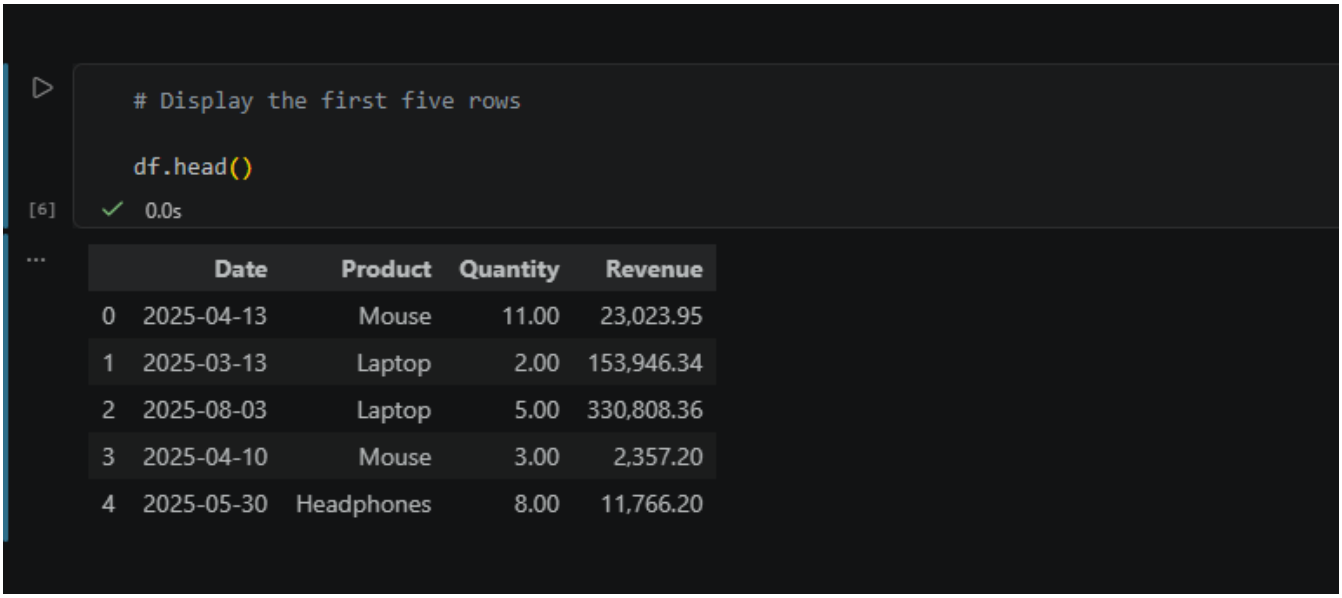
- Laptop is the highest revenue-generating product with ₹12,164,760.73.
- Smartphone is the second-highest revenue-generating product with ₹6,885,490.38.
- Mouse has the highest sales volume at 405 units.
- Mouse is also the lowest-revenue product, generating ₹607,344.02.
- High unit volume does not necessarily mean high revenue.

- May 2025 is the highest-revenue month with ₹3,511,756.91.
- November 2025 is the lowest-revenue month with ₹1,009,732.88.
- Laptop contributes approximately half of total revenue, making it a critical product.
- Monthly revenue varies considerably across the year.

13. Screenshots

This section contains the actual project-execution screenshots supplied for the submission. They document data loading, missing-value analysis, cleaning, product aggregation, visualizations, business insights, and final results. The screenshots are presented in execution order.

Screenshot 1 — Original Dataset / Initial Inspection



	Date	Product	Quantity	Revenue
0	2025-04-13	Mouse	11.00	23,023.95
1	2025-03-13	Laptop	2.00	153,946.34
2	2025-08-03	Laptop	5.00	330,808.36
3	2025-04-10	Mouse	3.00	2,357.20
4	2025-05-30	Headphones	8.00	11,766.20

Figure 13.1. Original Dataset / Initial Inspection.

Screenshot 2 — Missing Value Analysis

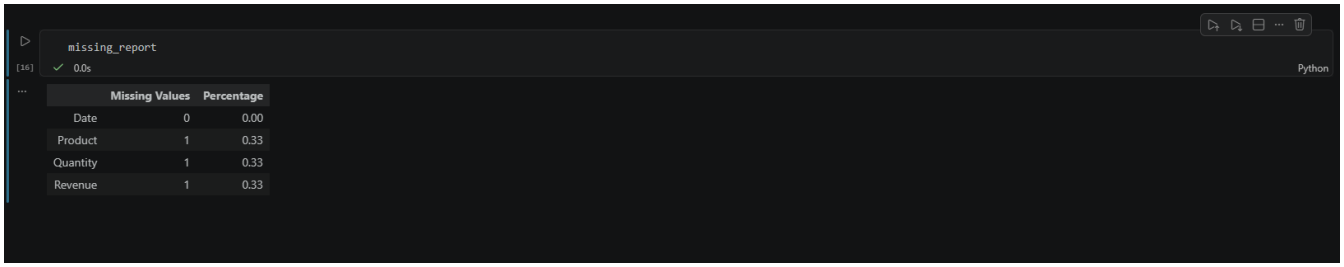
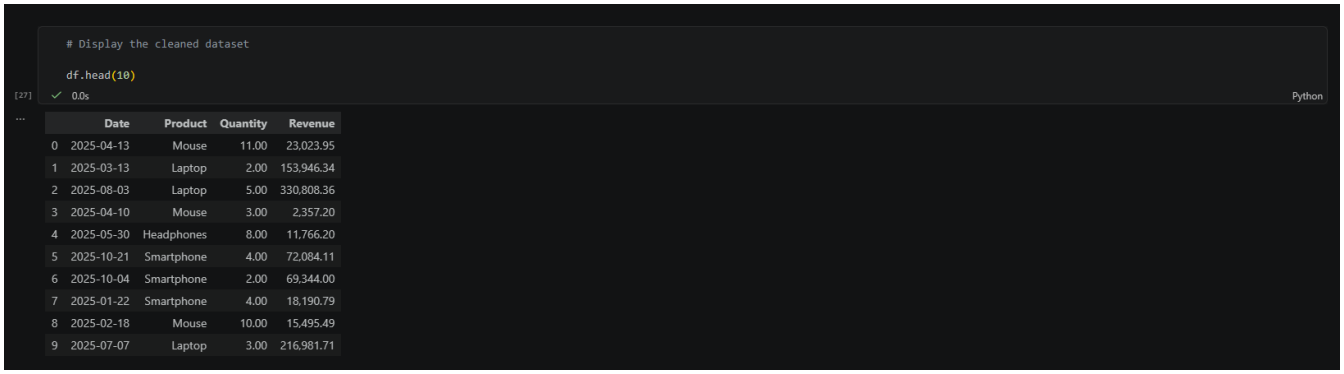


Figure 13.2. Missing Value Analysis.

Screenshot 3 — Cleaned Dataset



```
# Display the cleaned dataset

df.head(10)
```

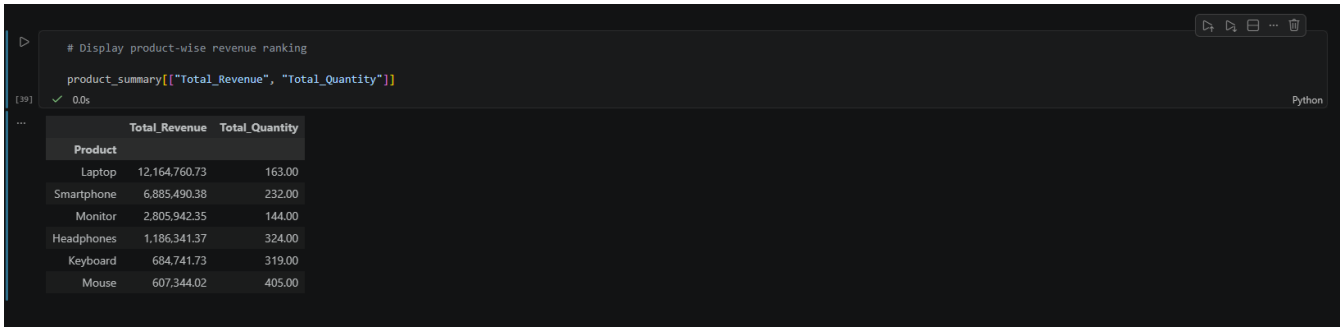
[27] ✓ 0.0s Python

...

	Date	Product	Quantity	Revenue
0	2025-04-13	Mouse	11.00	23,023.95
1	2025-03-13	Laptop	2.00	153,946.34
2	2025-08-03	Laptop	5.00	330,808.36
3	2025-04-10	Mouse	3.00	2,357.20
4	2025-05-30	Headphones	8.00	11,766.20
5	2025-10-21	Smartphone	4.00	72,084.11
6	2025-10-04	Smartphone	2.00	69,344.00
7	2025-01-22	Smartphone	4.00	18,190.79
8	2025-02-18	Mouse	10.00	15,495.49
9	2025-07-07	Laptop	3.00	216,981.71

Figure 13.3. Cleaned Dataset.

Screenshot 4 — Product-wise Analysis



```
# Display product-wise revenue ranking

product_summary[["Total_Revenue", "Total_Quantity"]]
```

[39] ✓ 0.0s Python

Product	Total_Revenue	Total_Quantity
Laptop	12,164,760.73	163.00
Smartphone	6,885,490.38	232.00
Monitor	2,805,942.35	144.00
Headphones	1,186,341.37	324.00
Keyboard	684,741.73	319.00
Mouse	607,344.02	405.00

Figure 13.4. Product-wise Analysis.

Screenshot 5 — Revenue by Product Bar Chart

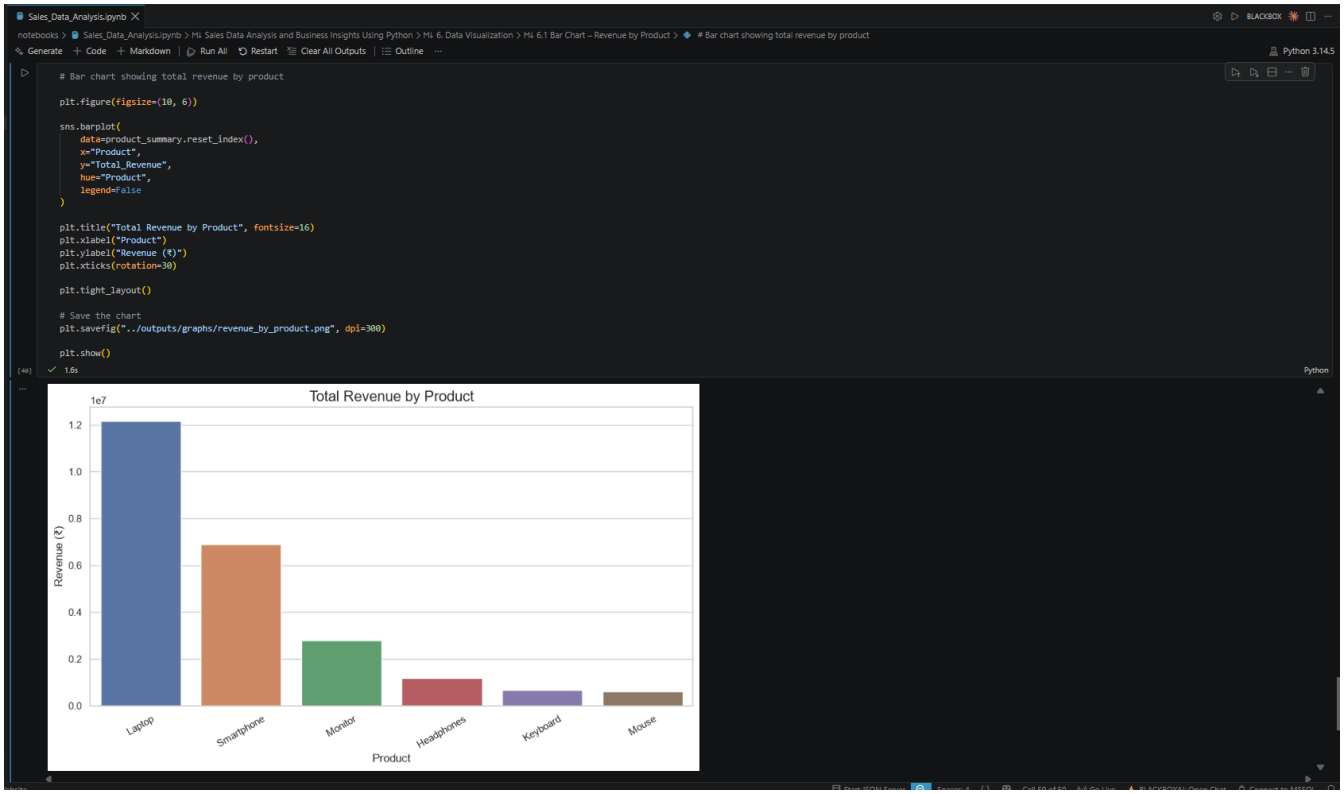


Figure 13.5. Revenue by Product Bar Chart.

Screenshot 6 — Monthly Revenue Trend

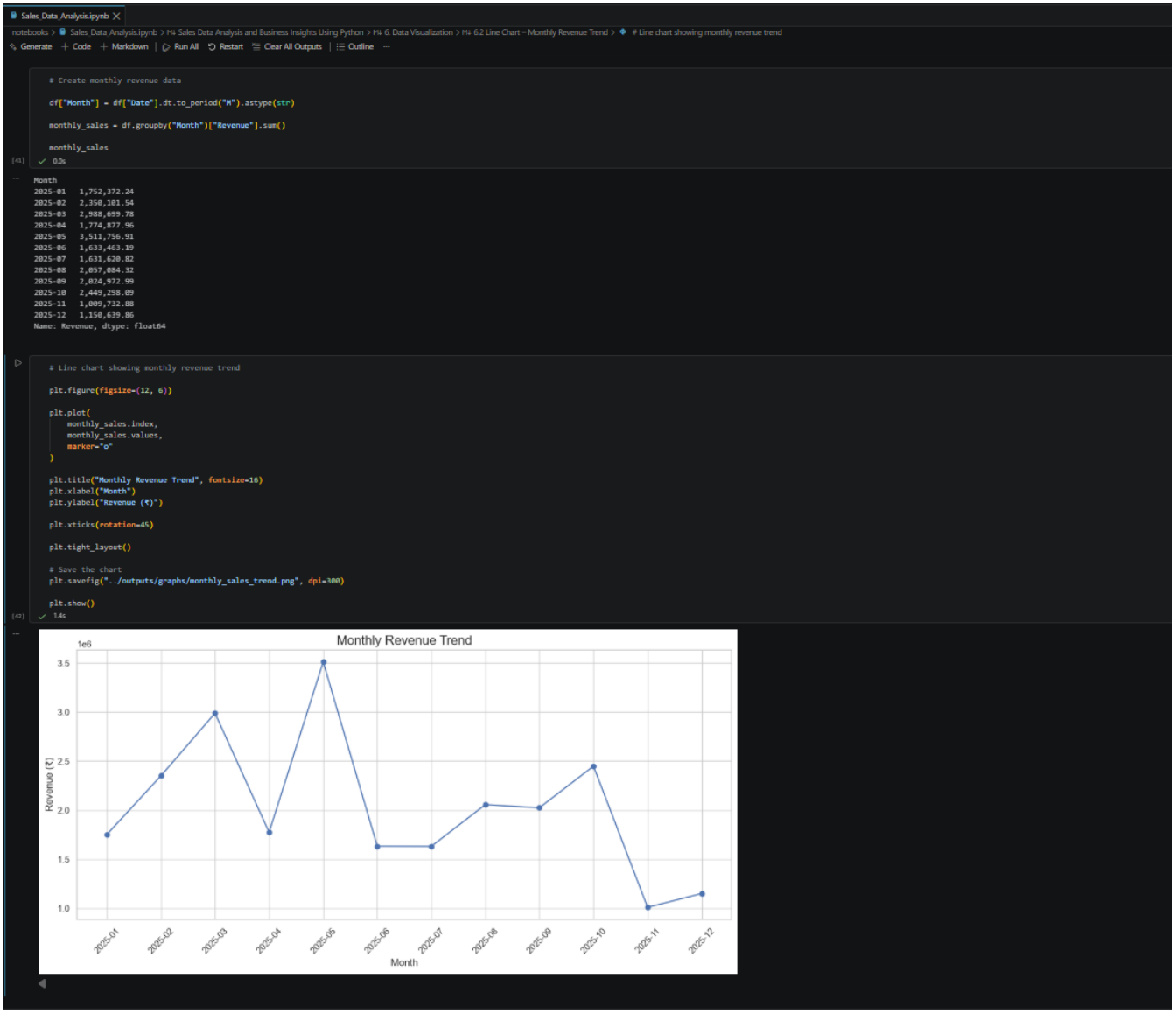


Figure 13.6. Monthly Revenue Trend.

Screenshot 7 — Revenue Share by Product

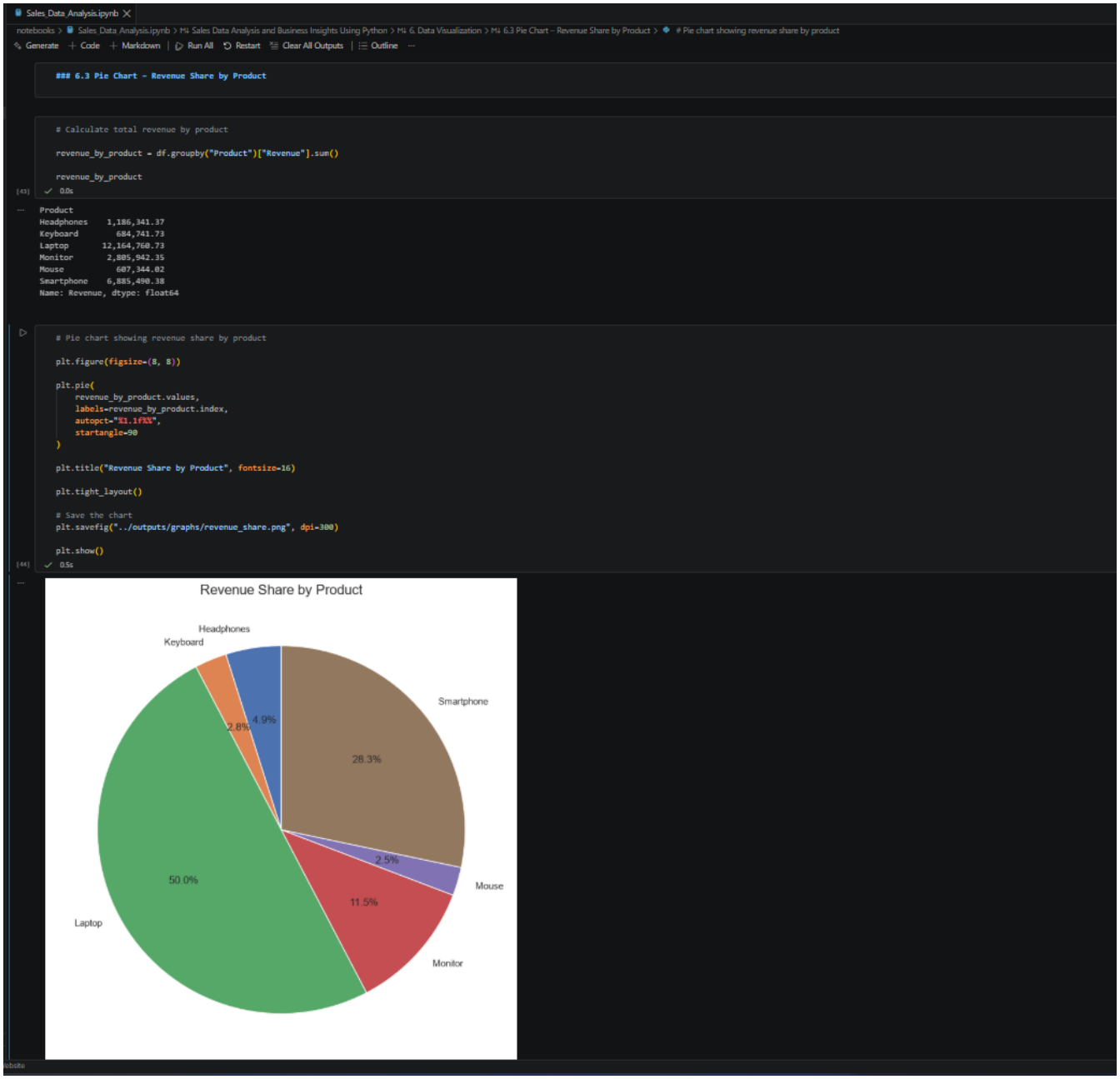


Figure 13.7. Revenue Share by Product.

Screenshot 8 — Business Insights

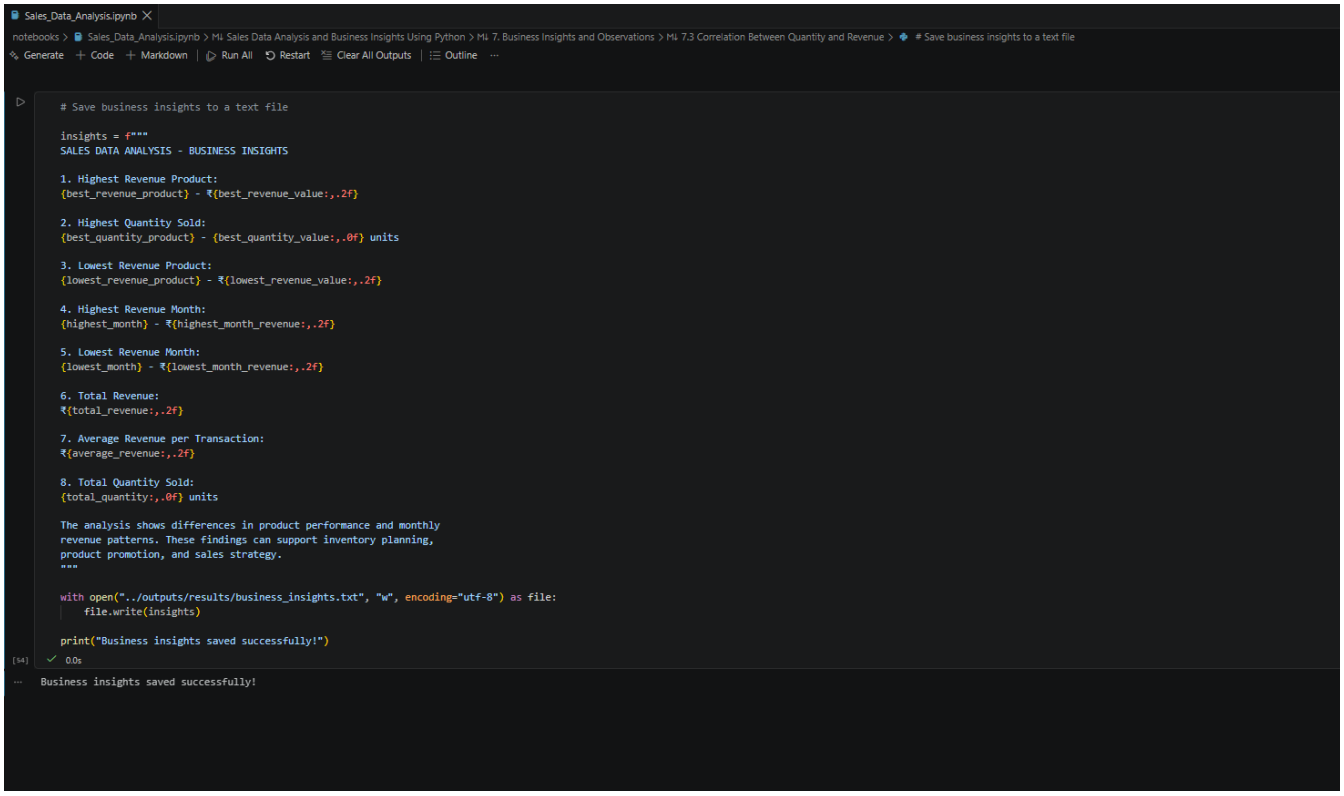


Figure 13.8. Business Insights.

Screenshot 9 — Final Project Results

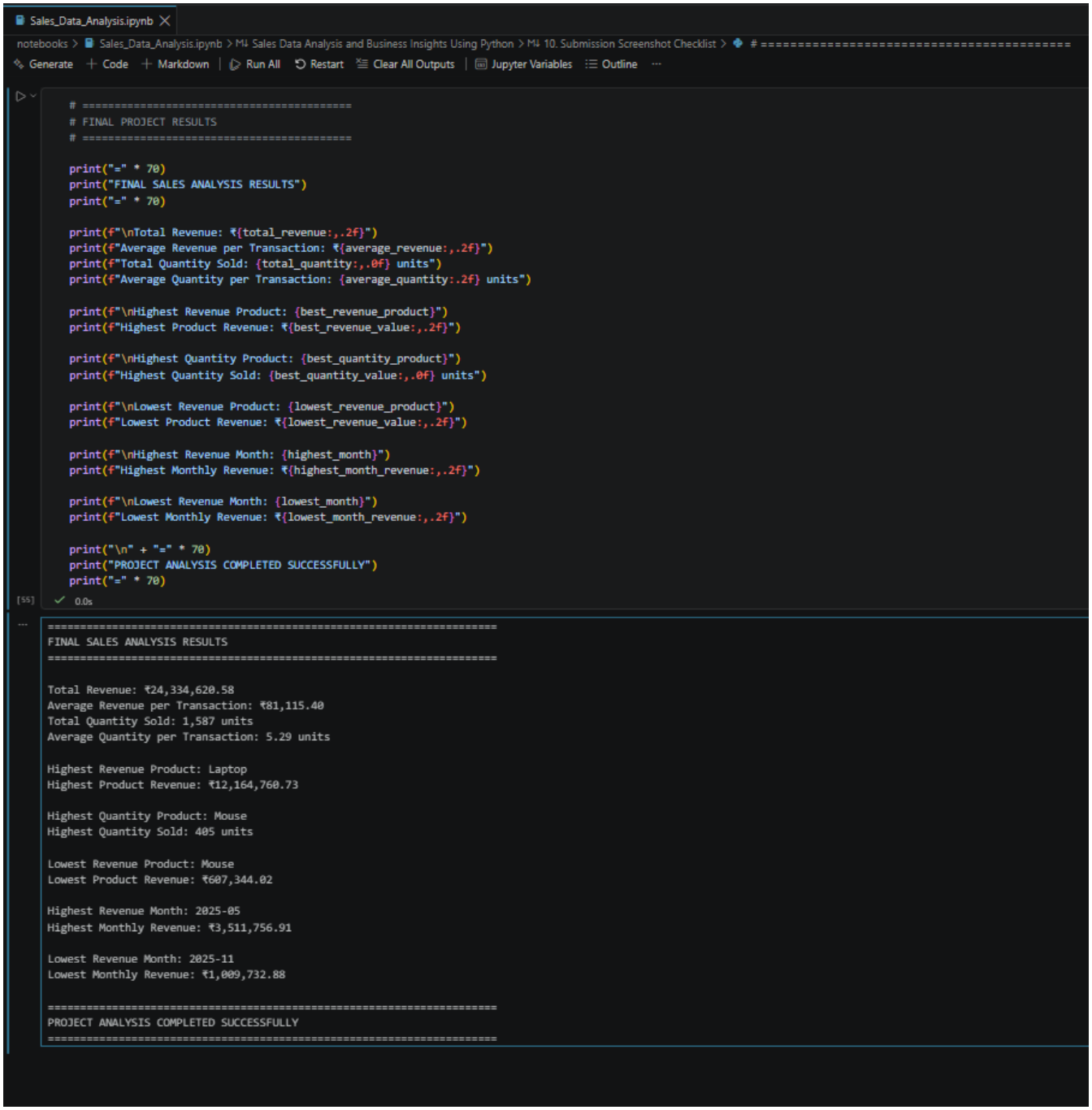


Figure 13.9. Final Project Results.

14. Final Results

Metric	Result
Total Revenue	₹24,334,620.58
Average Revenue per Transaction	₹81,115.40
Total Quantity Sold	1,587 units
Average Quantity per Transaction	5.29 units
Highest Revenue Product	Laptop
Highest Product Revenue	₹12,164,760.73
Highest Quantity Product	Mouse
Highest Quantity Sold	405 units
Lowest Revenue Product	Mouse
Lowest Product Revenue	₹607,344.02
Highest Revenue Month	May 2025
Highest Monthly Revenue	₹3,511,756.91
Lowest Revenue Month	November 2025
Lowest Monthly Revenue	₹1,009,732.88

15. Conclusion

The Exploratory Data Analysis project was completed successfully using Python and standard data science libraries. The workflow covered dataset loading, missing-value analysis, data cleaning, product-wise aggregation, monthly trend analysis, visualization, and business interpretation. The results demonstrate that Laptop is the strongest contributor to revenue, Mouse has the highest quantity sold, May 2025 is the strongest month, and November 2025 is the weakest month.

16. Future Scope

- Build an interactive Streamlit sales dashboard.
- Add profit, cost, and margin analysis.
- Perform sales forecasting using time-series models.
- Perform customer-level analysis when customer data is available.
- Detect unusual transactions and revenue anomalies.
- Study seasonal patterns and year-over-year growth.
- Use machine learning for demand prediction and inventory planning.

17. Screenshot Checklist

No.	Required Screenshot / Evidence	Status
1	Original dataset / initial inspection	Included

2	Missing-value analysis	Included
3	Cleaned dataset	Included
4	Product-wise analysis	Included
5	Bar chart — revenue by product	Included
6	Line chart — monthly revenue trend	Included
7	Pie chart — revenue share by product	Included
8	Business insights	Included
9	Final project results	Included

The supplied evidence contains 9 screenshots. They are included in Section 13 in the same order as the checklist above.

18. References

- Python documentation.
- Pandas documentation.
- NumPy documentation.
- Matplotlib documentation.
- Seaborn documentation.
- The numerical results and conclusions in this report are based on the completed project dataset, notebook execution, and screenshots supplied for this Minor Project.